Application of a GRU-Based Deep Learning Model to Environmental Monitoring at Complex Sites – 24471

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ABSTRACT

Prediction of the long-term fate of contaminants released to the subsurface is challenging. In traditional, physics-based models, it can be difficult to predict transport of contaminants in the subsurface due to variability in the conditions and sediments which may be unknown. However, with artificial intelligence (AI) based methods, such as deep learning, we can identify complex relationships with parameters that may not be clear from the available data. These methods can be applied in a wide range of applications for accurate predictions while dealing with multiple input parameters. Here, we apply AI-based methods for prediction of the fate of hexavalent chromium (Cr(VI)) in the subsurface. Cr(VI) is a hazardous, carcinogenic chemical that was historically released to groundwater at the Hanford Site during production and chemical processing operations for plutonium. The popular sequential deep learning model, long-short-term memory (LSTM), has been used for forecasting Cr(VI) concentrations in groundwater, but prediction accuracy is limited when considering multiple input parameters with a limited number of training samples.

To improve forecasting accuracy, in this research, we propose a framework that uses a gated recurrent unit (GRU) deep learning model to forecast Cr(VI) given multiple input parameters. GRU is widely used in many time-series applications to model non-linear relationships between multiple input parameters, especially when the data is limited for training the model. To conduct this experiment, time-series data for Cr(VI), groundwater temperature, and well water levels were obtained from the PHOENIX website with observations from different wells in the Hanford 100 Area. Weather data was also collected from the Hanford Meteorological Station. The data was then pre-processed and transformed into samples suitable for training the proposed model. The performance of GRU was compared with LSTM based on different evaluation metrics, such as mean squared error, root mean squared error, and coefficient of determination. The results showed that forecasting accuracy was improved by using GRU compared with LSTM. By understanding the long-term pattern of Cr(VI) at the Hanford Site based on multiple other parameters, the proposed approach may enhance Cr(VI) forecasting accuracy which could help to optimize future monitoring efforts across the Site.

INTRODUCTION

Predicting the effect of contaminants released into the subsurface poses a significant challenge. Traditional approaches to environmental monitoring and forecasting often rely on classical models [1]–[3], which are effective with simple systems and pose significant limitations when applied to multifaceted data [4]. The contaminants' behavior in the subsurface is complex due to other unknown parameters in the environment, which makes predictive modelling challenging. Recent advances in artificial intelligence (AI) through deep learning are revolutionizing data analysis across a wide range of domains, providing robust solutions to modeling non-linear relationships that exist in the datasets for predicting contaminant fate. For example, Long-term heavy metal concentrations in aquatic ecosystems is modelled using artificial neural network in [6]. In this study, we apply AI-based solutions to predict hexavalent chromium (Cr(VI)) concentrations in

the subsurface. Cr(VI) is a dangerous pollutant, known for causing cancer and other health issues, commonly released from industrial processes into the subsurface environment. In general, Cr(VI) concentrations in groundwater are influenced by a confluence of factors [5], including groundwater temperature, weather conditions, and water level. The relationships between these parameters can be challenging to understand in complex systems with a range of conditions and variables.

Deep learning-based sequential models, like Long Short-Term Memory (LSTM) [7], have proven effective for forecasting Cr(VI) at the Hanford Site. However, the limited number of training samples has constrained the potential of LSTM models, particularly when dealing with multiple input parameters, as shown in our experiments. To overcome this limitation, alternative deep learning architectures must be explored that can better capture environmental dynamics with fewer samples and multiple parameters.

In this study, we propose a framework that uses a gated recurrent unit (GRU) [8] based deep learning model to forecast Cr(VI) concentrations at the Hanford Site based on multiple input parameters. Due to its capability to handle time-series data, the GRU is used for forecasting various events in different applications [9]–[12]. It is particularly suitable for applications that deal with limited data [13], [14] for training the model. Compared to its LSTM counterpart, GRU simplifies and optimizes learning processes, which can be crucial when modeling the complex interactions between multiple environmental factors affecting Cr(VI).

BACKGROUND

The ability of deep learning algorithms [15] to model non-linear relationships between time-series input parameters accelerate AI applications in different domains for forecasting future trends given past observations. Some of the notable time-series applications are stock market forecast [16], [17], weather forecast [18], [19], disease spread forecast [20], [21], electricity load forecasting [22], air quality prediction [23], etc. In general, forecasting a parameter may be influenced by more than one parameter. Therefore, it is important to consider multiple input parameters for forecasting. However, it increases model complexity [4] to extract non-linear patterns from input parameters. In addition, when the number of samples is limited for training the model, variance increases [24] in the model forecast, resulting in sub-optimal predictions. This section presents a set of related works on different models handling multiple input parameters and/or with limited training samples in time-series applications.

Multiple input parameters for time-series forecasting: To forecast water quality, the authors in [4] consider ammonia and dissolved oxygen parameters that exhibit a complex relationship. Therefore, a deep learning-based ensemble method is utilized to extract non-linear patterns between the input parameters. In [25], cascade-forward networks are used to predict water quality by analyzing nonlinear relationships between the water properties, chemical properties, and flow status parameters. In certain applications, time-series data may not hold seasonal trends while considering multiple input parameters. Consequently, [26] proposes simple hidden Markov models based on fuzzy rules. The fuzzy inference engine parameters are optimized using particle swarm optimization. In [27], snow cover and normalized difference vegetation index are taken as input and output using LSTM model. In [28], GRU is employed to process multiple input parameters and detect anomalies in the time-series data in industrial control systems. To monitor and control the sea surface water quality of the Karachi coastal area along the harbor, the authors in [29] apply a nonlinear auto-regressive exogenous neural network to model complex relationships between a set of input and output parameters for forecasting heavy metal contamination. The physical parameters, such as sea

surface temperature, salinity, tides, and pH are considered input parameters and the chemical parameters chromium, copper, lead, nickel, and zinc are considered output parameters.

Dataset with limited samples in prediction: Although deep learning-based sequential models provide excellent performance on time-series data, their forecast power becomes suboptimal when there are not enough training samples. The model variance increases in this situation, resulting in unreliable predictions of future trends. In [14], the authors propose a modified GRU to deal with a small number of samples for predicting sign language. In this method, temporal patterns are extracted from sequences of image frames. The results and analysis of [30] also recommend GRU for applications with small datasets and bi-GRU for applications with large datasets. GRU is extensively analyzed in [31] to demonstrate its power over LSTM when compared to limited datasets. Similar analysis of time-series forecasting is discussed in [32].

PROPOSED METHODOLOGY

This section provides detailed information on the different modules in the proposed framework, as shown in Figure 1, for developing and refining a GRU model for forecasting Cr(VI) concentrations, emphasizing data collection and pre-processing, model development, and real-time prediction.

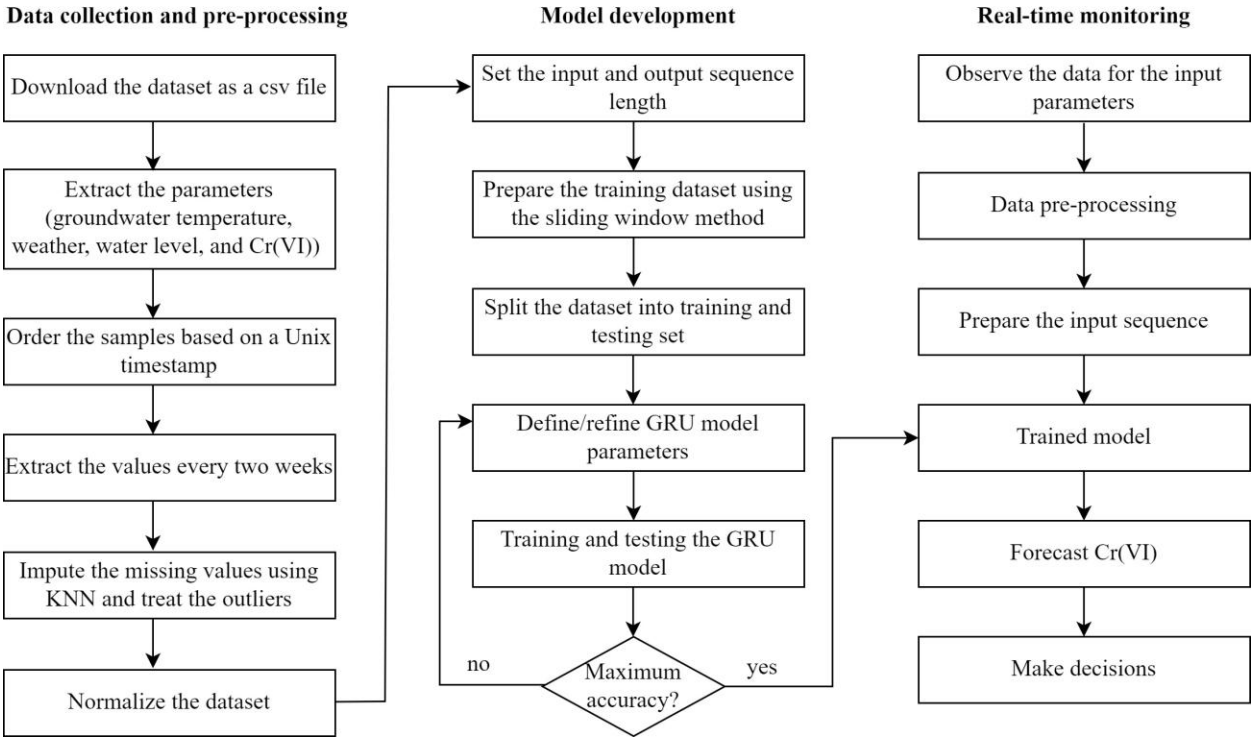


Figure 1. Steps in the proposed framework

Data Collection and Pre-Processing

Downloading the Dataset: The dataset, obtained in CSV format, is sourced from the PHOENIX database [33] managed by Pacific Northwest National Laboratory (PNNL) and the Hanford Meteorological Station [34]. This dataset contains a historical record of environmental parameters.

Parameter Extraction: The dataset includes various parameters recorded over different periods. Therefore, we need to extract the parameters that are needed for the experiment. For this study, specific data are

extracted: Cr(VI) concentration, groundwater temperature, and well water level from different wells in the Hanford Site's 100-area, and weather data from the Hanford Meteorological Station. In the proposed framework, groundwater temperature (°C), well water level (meter), Cr(VI) ($\mu g/L$), and weather (°C) serve as input parameters, with Cr(VI) ($\mu g/L$) also being the target output parameter.

Ordering the Samples by Time: The available data spans 1969 to 2023. However, the observed samples in the database are not arranged by time. To address this issue, dates are converted into Unix timestamp format, which offers a unique and consistent numerical representation of time, with each second since January 1, 1970. With this conversion time can be represented in a standardized way, allowing observations for each parameter to be sorted based on their timestamps. Consequently, this reordering ensures that the dataset is aligned in time, which helps for subsequent stages of data processing and analysis in GRU model development.

Bi-weekly Data Extraction: Data for all relevant parameters is extracted for the period 2000 to 2023, at two-week intervals. This approach was chosen due to the asynchronous nature of data recording across different parameters. This resulted in 552 distinct data samples.

Imputation of Missing Data: Despite the two-week interval strategy, the dataset exhibits missing values. The number of missing values is less than 2% of the biweekly extracted dataset. As the number of observations is in the few hundreds, removing the missing samples will reduce the quality of the model. Therefore, rather than discarding these incomplete records, missing values are imputed using the K-nearest neighbour (KNN) method [35], preserving the dataset's integrity and size. Imputing missing values using KNN is done based on similarity. By using a distance metric like Euclidean distance, the KNN algorithm identifies 'k' nearest neighbours (k=10 in our method) when a missing value is encountered. As a result, we can estimate the missing value based on the context provided by these neighbours. Based on these neighbours, mean is calculated to impute the missing value. KNN-based imputation maintains the variance and structure of the data, which is very important for training the GRU model effectively, especially considering the non-linear relationships among environmental parameters like Cr(VI) levels, groundwater temperature, and weather conditions.

Outlier Treatment: To ensure data quality and model accuracy, outliers are identified and treated. Data points with a Z-score [36] beyond the +3 or -3 range are considered outliers and removed. The gaps created by this removal are filled using the KNN imputation method. Outliers in all parameters are treated using this method.

Normalization of Dataset: In this research, the parameters, such as temperature is measured in C, well water level is measured in meters, and Cr(VI) concentration is measured in ug/L. When working with input parameters on different scales, the model may incorrectly interpret larger-scaled features as having higher importance. Therefore, the parameters are normalized to the common scale, without affecting differences in value ranges or losing information. We use the Standard Scaler [37] from the Scikit-Learn Python library to achieve this goal. The Standard Scaler standardizes the features by removing the mean and scaling to unit variance. This process transforms each feature to have a mean of zero and a standard deviation of one, thus adjusting the different scales of the input parameters to a uniform range. This step is very important for the efficient performance of the GRU model, as it ensures that all parameters contribute equally to the model's learning process, preventing any bias towards a particular feature due to its original scale.

Model Development

Setting Input and Output Sequence Lengths: Sequential models (like LSTM and GRU) require a defined sequence length for both input and output. In this case, the sequence length is set to 4 for both input and output. However, it is a hyperparameter that depends on the complexity of the parameter relationships and the dataset size.

Preparation of a Training Dataset: The sliding window method [38] is employed to generate the dataset for model training. This method ensures that sequential dependencies within the data are adequately captured. Figure 2 shows how a sliding window is applied to the time-series data of n observations of all the parameters. The red color window indicates the input sequence of values for the parameters and the blue color window indicates the output sequence of values. Figure 2 shows two training samples. Until the last observation, the windows are moved in this manner to prepare training samples. As a result of preparing samples like this, the time series forecasting problem becomes a supervised learning problem.

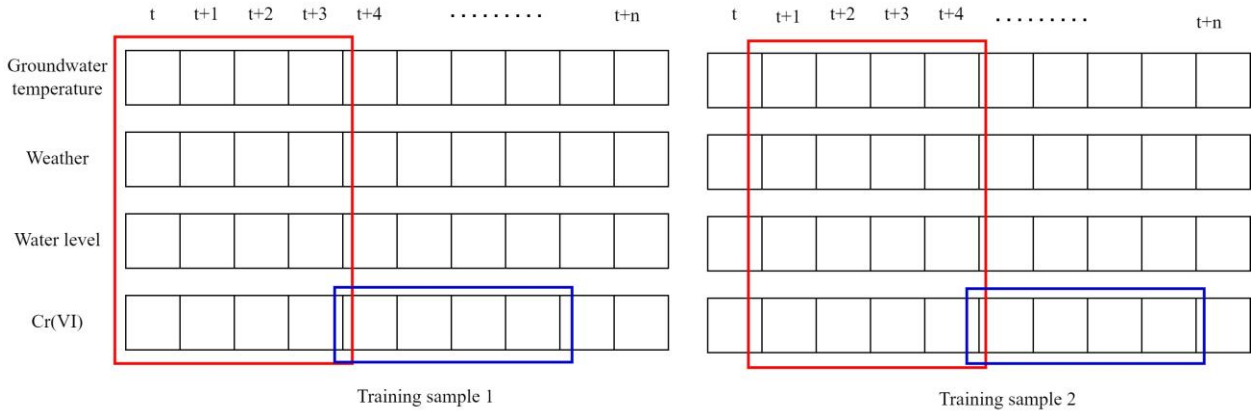


Figure 2. Training data preparation

Dataset Splitting: The prepared dataset is split into training and testing sets, following an 80:20 ratio. This results in 438 samples for training and 110 for testing. The slight reduction in total sample size (from 552 to 548) is due to the sliding window method and its requirement for sequence length.

Define and Refine GRU Model and Parameters: The GRU model [8] comprises two parts: sequential layers and fully connected dense layers. The sequential layers process the input sequence of Cr(VI), groundwater temperature, weather, and well water level, learning inherent temporal patterns. These patterns are then fed into dense layers mapping to the output Cr(VI) sequence. The proposed GRU model (as shown in Figure 3) includes

- four neurons in the input layer, as there are four input parameters.
- two sequential layers with 128 and 64 neurons, respectively.
- a dense layer with 64 neurons.
- an output layer with 4 neurons (as the output sequence is 4).

The recurrent loop in the sequential layers is executed 4 times (as the input sequence length is 4) for every input sample, and the result is passed on to the dense layer.

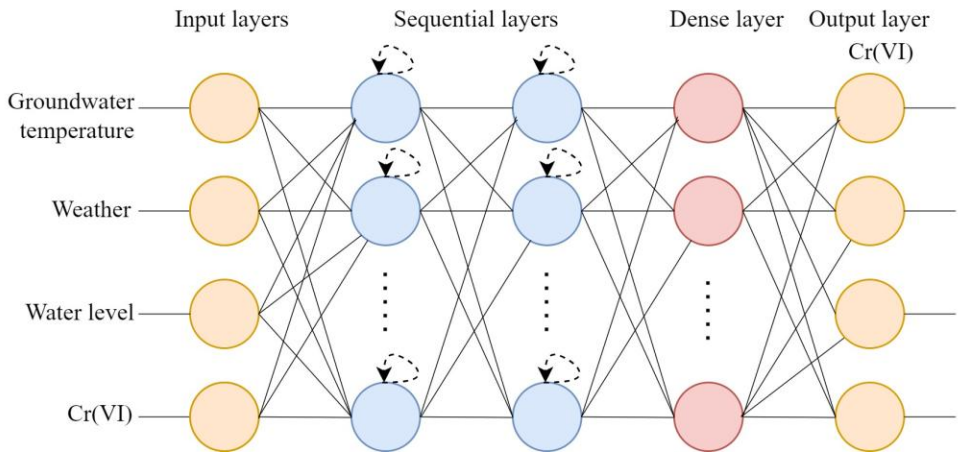


Figure 3. Layers in proposed GRU model

GRUs provide better results in scenarios where shorter sequences are present, as they can help reduce overfitting due to their simpler architecture. Unlike LSTM that takes three gates (forget, input, and update), every neuron in GRU sequential layers comprises only two gates (update and reset) for regulating information processing, as shown in Figure 4. The specifics are given below.

- x_t is the input of a GRU neuron.
- h_t is the current hidden state (output) of the neuron during recurrence.
- Update gate (z_t) determines what percentage of dependency in a sequence to retain from previous state (h_{t-1}) of the neuron.
- Reset gate (r_t) determines what percentage of dependency information to remove in a sequence.
- Candidate solution ($\tilde{h}_t$) is generated to determine how much percentage of current input sequence pattern to consider for updating with h_t.
- $\hat{y}_t$ Produces output to the neuron in the next layer.

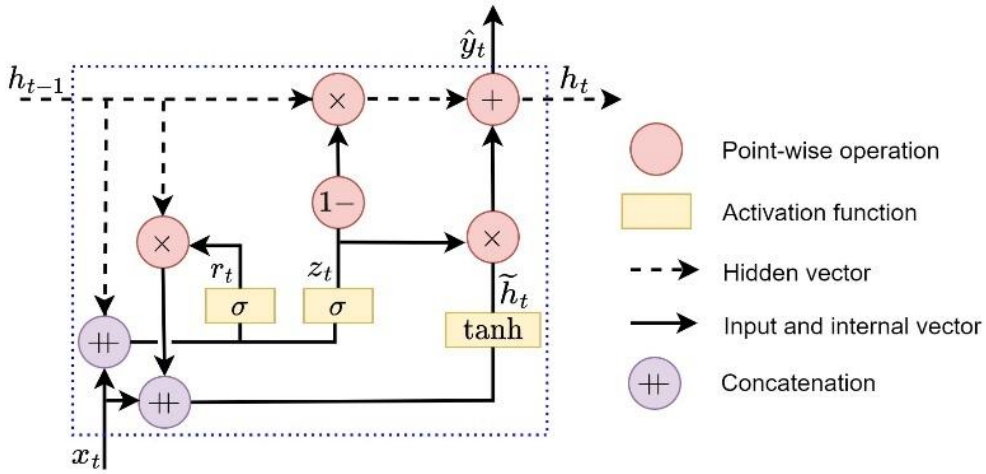


Figure 4. GRU cell (neuron) in sequential layers

The activation function used in the sequential layers are Sigmoid (σ) and TanH. ReLU is used in dense layers. Adam optimizer is used to improve the convergence speed of the mode with a batch size of 6 and 50 training epochs. LSTM is similarly configured for comparative performance analysis.

Training and Testing the GRU Model: After the GRU model is defined, it is trained using the training dataset and evaluated against the testing set.

Model Accuracy and Refinement: If accuracy is not improved, the model is redefined by tuning the hyperparameter values of the model and retrained. Hyperparameters, including input and output sequence lengths, optimizer, activation function, etc., are tuned and the model is retrained to achieve higher accuracy.

Real-time monitoring

Observing Data for Input Parameters: The parameters extracted from the data include: Cr(VI) concentration, groundwater temperature, weather, and well water level as observed at the Handford Site.

Data Pre-processing: The data is sorted using a Unix timestamp and the missing values (if any) are imputed by the KNN method, ensuring no loss of data integrity, based on a biweekly sampling. Additionally, outliers are identified and removed, with their corresponding gaps filled using KNN. The data is then normalized using the Standard Scaler.

Preparing the Input Sequence: The observed input data is transformed into a sequence of 4 values suitable for the GRU trained model.

Testing the Model: Once the GRU model is trained, predictions are compared to the testing dataset. The primary objective of the model is to forecast Cr(VI) concentrations accurately in groundwater. The model, trained on historical data, uses its learned patterns to predict future Cr(VI) levels.

Forecasting: The pre-processed data is passed onto the model for forecasting Cr(VI).

Making Decisions: Forecasting Cr(VI) is crucial for understanding potential future trends in contamination levels. The predictions provided by the GRU model could serve as a guide for environmental scientists and decision-makers at the Hanford Site. Based on these forecasts, proactive remediation strategies could be developed to manage and mitigate Cr(VI) contamination.

RESULTS

The results are presented in three sections: performance analysis, forecasting Cr(VI), and prediction and confidence interval analysis.

Performance analysis

The proposed framework with the GRU model is evaluated with Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared (R^2) value for both training and testing datasets. Table 1 compares the performance of GRU with LSTM in both single and multiple variable settings.

Single input parameter: The LSTM model shows moderate effectiveness in predicting future Cr(VI) using only historical Cr(VI) data. In this case, given a sequence of Cr(VI) values, the first 4 values are considered as input and the next 4 values are considered as output. With LSTM, the higher RMSE on the test data compared to the training data indicates a decrease in performance when generalizing to unseen data. The

GRU model demonstrates a similar performance to LSTM on the training data but shows a slightly better generalization on the test data. The improvement in R^2 value from 83.7 to 86.7 indicates that the GRU model can explain a higher percentage of the variance in the target variable. It is important to note, however, that forecasting Cr(VI) is limited, regardless of the model used with one input parameter.

Table 1. Performance metrics

Model	Training data		Test data		R^2
	MSE	RMSE	MSE	RMSE	
LSTM (Univariate)	150.15	12.25	188.5	13.73	83.7
GRU (Univariate)	149	12.21	186.35	13.65	86.7
LSTM (Multivariate)	34.48	5.87	100.68	10.03	95.2
GRU (Multivariate)	30.19	5.49	93.33	9.66	98.5

Multiple input parameters: To improve predictive accuracy, additional parameters to Cr(VI), such as groundwater temperature, weather, and groundwater level are added to the model. As shown in Figure 2, samples for training and testing are prepared with an input and output sequence length of 4. LSTM performs better in this scenario than its univariate counterpart. Lower RMSE values indicate better accuracy. However, the difference between training and testing RMSE suggests that there exists some overfitting. It is also observed that adding multiple parameters to forecast Cr(VI) using GRU model improves the predictive accuracy up to 12% compared to the GRU model with single parameter. In addition, the GRU model outperforms LSTM as well, both in training and testing data. The GRU model yields lower error rates and exhibits a higher R^2 value of 98.5, with a 3.3% improvement compared to LSTM. This suggests that the GRU model can explain a significantly higher proportion of output parameter variance. It is because GRU's architecture allows it to learn and adapt faster than LSTM from fewer samples with multiple parameters.

Forecasting Cr(VI)

Figure 5 (a) and (b) shows the forecasting performance LSTM and GRU models with multiple parameters for different samples in the test set. The GRU model forecasts the sequence (next 4 biweekly Cr(VI) concentrations) given the past 4 observed values for all the input parameters. The predictions are much closer to the test set compared to the LSTM predictions.

Prediction and confidence interval

Prediction intervals estimate the range within which future observations are likely to fall, while confidence intervals provide a range that likely contains the true value of a model parameter. To calculate these intervals, we fit a regression on the observed and predicted Cr(VI) value for both LSTM and GRU. As shown in Figures 6 and 7, the prediction and confidence intervals for the GRU model are observed to be significantly tighter and closer to the regression line compared to those of the LSTM model. The narrower intervals suggest that the GRU model has a stronger certainty in its forecasts, with less variability and potential error. In contrast, the wider intervals observed with the LSTM model imply increased uncertainty and higher variance. Furthermore, this observation aligns with the GRU model's lower error rates and higher

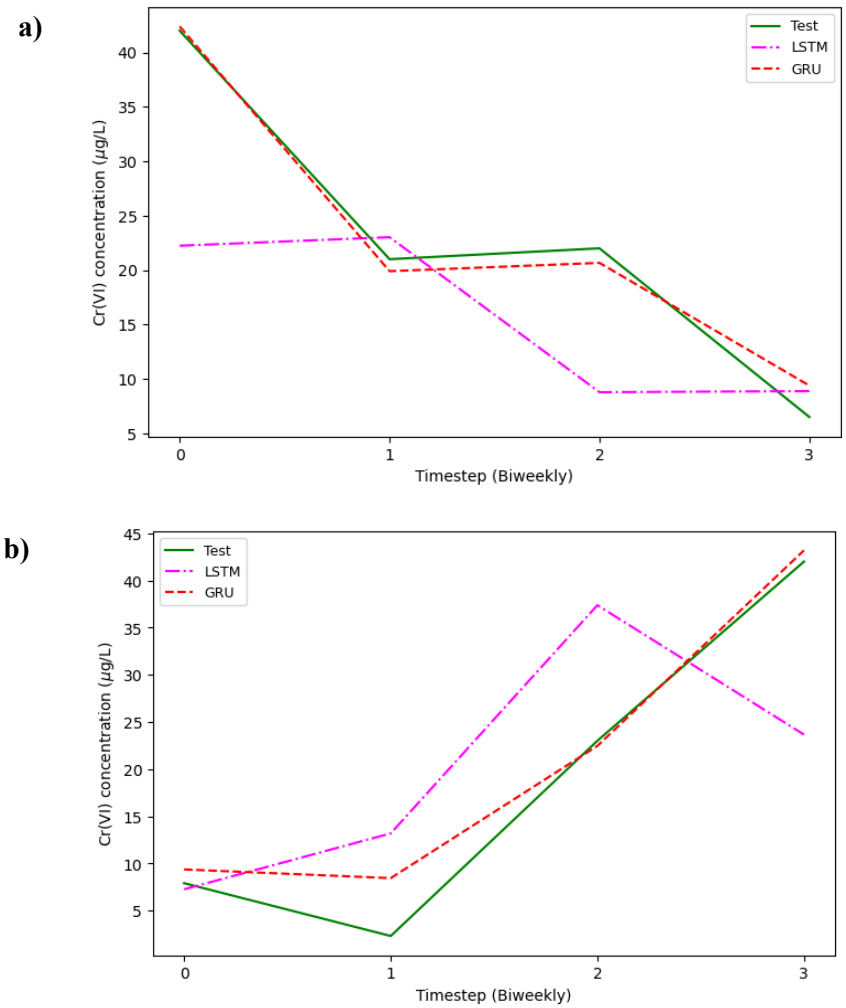


Figure 5. Forecasted Cr(VI) with different samples in the test set: (a) instance 1 (b) instance 2

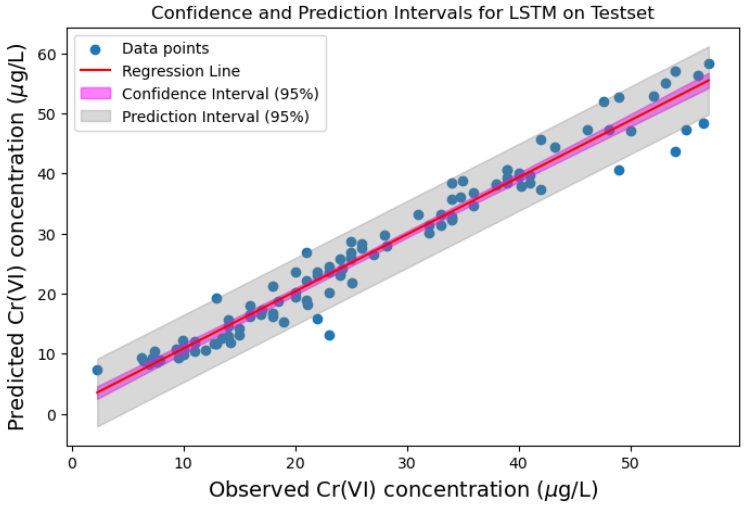


Figure 6. Confidence and prediction intervals for LSTM with multiple input parameters on the test

R^2 values. It suggests that the GRU model is not only effective at making accurate Cr(VI) predictions but also provides reliable uncertainty estimates.

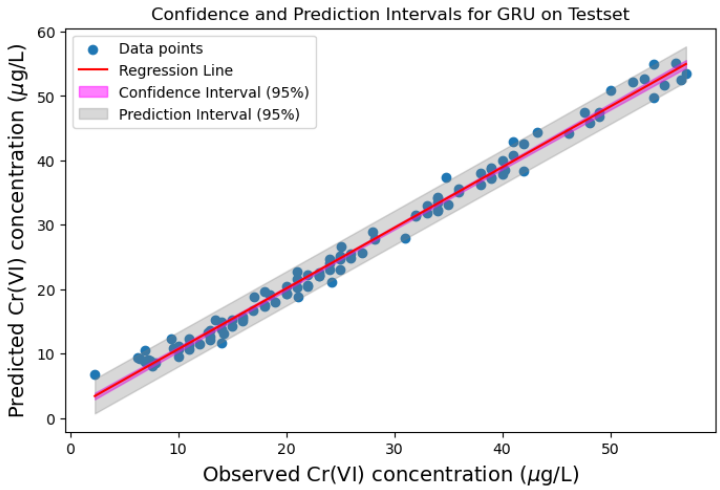


Figure 7. Confidence and prediction intervals for GRU with multiple input parameters on the test

CONCLUSION

We proposed a framework with a GRU model, which is suitable for applications with a lesser number of samples for training the model, for forecasting contaminant concentrations in the subsurface based on historical monitoring data. The results showed that predictive accuracy is improved when using multiple input parameters (e.g., contaminant concentration, pH, temperature) compared to a model with a single input parameter (e.g., contaminant concentration). In addition, GRU performance was compared with LSTM in similar settings. GRU model significantly outperformed LSTM in forecasting Cr(VI) concentrations in groundwater at the Hanford Site.

This research underscores the potential of GRU models in environmental monitoring and management, providing a robust tool for effective prediction of contaminant concentrations potentially enabling more efficient monitoring and remediation planning for subsurface environmental cleanup activities. Accurately forecasting groundwater Cr(VI) concentrations at contaminated sites like the Hanford Site is crucial for developing effective remediation strategies. This research demonstrates that considering multiple parameters (groundwater temperature, weather, and well water level) to forecast Cr(VI) improves prediction accuracy.

In the future, real-time data integration for dynamic model updating could significantly improve the model's performance in response to environmental changes. To broaden this research work's applicability to environmental management, the model could also be explored for scalability to other contaminated sites with different characteristics and challenges.

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